🔑 **Answer Key — Teacher Copy · Lesson 2-7**

MSTAR PRACTICE WORKSHEET

Lesson 2-7: Divide Decimals Using an Algorithm

Grade 6 Mathematics · Standard 6.NOS.3

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is 14.4 ÷ 1.2?

A. 0.12

B. 1.2

C. 12

D. 120

Multiply both by 10: 144 ÷ 12 = 12.

Question 2 SELECTED RESPONSE — CORRECT: C

What is 3.78 ÷ 0.6?

A. 0.063

B. 0.63

C. 6.3

D. 63

Multiply both by 10: 37.8 ÷ 6 = 6.3.

Question 3 SELECTED RESPONSE — CORRECT: C

What is 8.4 ÷ 0.7?

A 0.12

B 1.2

C 12

D 120

8.4 ÷ 0.7 = 84 ÷ 7 = 12.

Question 4 SELECTED RESPONSE — CORRECT: B

What is 15.6 ÷ 1.3?

A 1.2

B 12

C 13

D 120

Multiply both by 10: 156 ÷ 13 = 12.

Question 5 SELECTED RESPONSE — CORRECT: C

A water tank holds 19.2 liters. It is poured into bottles that each hold 1.6 liters. How many bottles can be filled?

A 1.2 bottles

B 11 bottles

C 12 bottles

D 13 bottles

19.2 ÷ 1.6: multiply both by 10 → 192 ÷ 16 = 12 bottles.

Question 6 SELECTED RESPONSE — CORRECT: C

What is 4.8 ÷ 0.6?

A 0.8

B 4.2

C 8

D 80

Multiply both by 10: 48 ÷ 6 = 8.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Find 18.9 ÷ 6.3 using the decimal division algorithm.

A 0.3

B 2.8

C 3

D 30

Slide both decimal points one place right: 189 ÷ 63 = 3. Multiplying both numbers by 10 leaves the quotient unchanged.

- **A:** The quotient was divided by 10 after the points were slid. Moving BOTH points the same direction leaves the answer unchanged.
- **B:** This division comes out even: 63 x 3 = 189, so the quotient is a whole number.
- **D:** The quotient was multiplied by 10 after the points were slid. 189 divided by 63 is exactly 3.

Part B — correct answer: D

Why does sliding BOTH decimal points the same number of places leave the quotient unchanged?

A The decimal point does not affect division.

B Only the divisor matters in a division problem.

C The quotient always doubles when you slide a point.

D Multiplying the dividend and the divisor by the same power of ten produces an equivalent division.

18.9 ÷ 6.3 and 189 ÷ 63 are equivalent because both numbers were multiplied by 10 — the same scaling on both terms of a ratio keeps its value.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Moving Only One Point

> Bianca computed 7.5 ÷ 0.5. She slid the divisor's point to make it 5, but left the dividend as 7.5, and wrote '7.5 ÷ 5 = 1.5.'

Explain Bianca's misconception and give the correct quotient.

Model answer: 75 ÷ 5 = 15. Bianca multiplied only the divisor by 10, which changes the problem. Both numbers must be scaled by the same factor, so 7.5 becomes 75. A check confirms it: 0.5 × 15 = 7.5.

2 points	Student explains that both decimal points must slide the same number of places to keep the division equivalent, computes 75 ÷ 5 = 15, and notes that 1.5 is ten times too small.
1 point	Student gives 15 without explaining why both points must move.
0 points	Student agrees with Bianca or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A gas station sells fuel at $3.45 per gallon. A customer pays $27.60 total. How many gallons did the customer buy? Set up the decimal division, show how to convert it to whole-number division, and solve.

Model answer: Total cost divided by price per gallon gives the number of gallons, so the setup is 27.60 ÷ 3.45. The divisor has two decimal places, so multiplying both numbers by 100 gives 2,760 ÷ 345 = 8 gallons, and 8 × $3.45 = $27.60 confirms it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.